Foundation for Virtual and Augmented Reality Systems

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Week – 02

Lecture – 11

Lec 11: Cybersickness

Hello and welcome to the NPTEL MOOCs course on Foundations of Virtual and Augmented Reality Systems. We are currently discussing cyber sickness as part of Lecture 6. So, we will continue our discussion on cyber sickness. So, in the previous part of the lecture, we have talked about, partly talked about the causes. Few factors that fall under two categories: one is internal, the other one is external hardware part.

Now, we are going to talk about the remaining category, that is the external content part. Also, in this part of the lecture, we will start our discussion on how to evaluate or measure cyber sickness. So, let us first quickly have a look and complete our discussion on the remaining part of the causes. So far, we have covered the internal factors: that is age, gender, motion sickness susceptibility, physical condition, and emotional condition. We have also partly covered the external factors.

More specifically, we have covered the hardware factors, hardware-related factors, namely display resolution, latency, latency jitter, type of display, and type of input. Now, we are going to talk about the factors related to the VR content. So, there are six such factors: visual motion, means of locomotion, field of view, type of content, duration, and visual complexity. So, let us go through each of these in some detail. The design of virtual environments can have a significant impact on the likelihood of CS, also the severity of CS.

One factor which is related to VR content is visual motion. Now, visual motion stimulus, which is the movement experienced by the user in a virtual environment, is considered to be one of the primary causes of cyber sickness. Higher such visual motion can increase the

risk of CS and The way the content moves also affects CS. The way the content moves within the virtual environment also affects CS and its severity.

Vection, or the sensation of self-motion, is the highest when we sense motion in peripheral stimuli. So, if there is motion in peripheral stimuli then our sensation of self motion or vection is the highest and it has been found that there is a positive correlation between vection and visual motion stimulus. So, that positive correlation affects

or indicates that visual motion plays a crucial role in the overall onset and severity of cyber sickness. In addition, the severity of CS also significantly increases with larger height and faster speed in the virtual environment. So, the viewpoint during motion is also an important factor. However, counter-intuitively, there is a finding that brief exposure to turbulent visual motion

is alone not enough to induce CS. So, typically we expect that if we are exposed to in the virtual world, we are exposed to turbulent motion, roller coaster ride sort of things, then that may lead to cyber sickness. However, brief exposure to such turbulent motions does not induce cyber sickness, but longer exposure certainly can and That is a somewhat counterintuitive finding. CS also increases with the increase in angular velocity and is more intense when the rotation is taking place around the x and y axes as compared to the z axis.

So, that is another finding from the studies. That is about visual motion and cyber sickness. Then another content-related factor is means of locomotion. How we are moving in the virtual world. A sensory mismatch can occur when users' virtual movements do not match their physical movements, and this mismatch results in discomfort and disorientation which

in turn leads to cyber sickness. One aspect of locomotion that affects cyber sickness is control over motion. Whether we are controlling the motion and in which manner that affects cyber sickness. That control over motion has a direct effect on presence and an indirect effect on cyber sickness through presence. A greater sense of presence in virtual reality can lead to a lower likelihood of experiencing CS in general.

Of course, it need not always hold true, but in general, greater presence leads to lesser CS and And control over locomotion is related to presence. More control means greater presence; less control means lower presence. Now, the type of control mechanism used in VR, as well as the extent of control. So, the type of control and the extent of control are both important.

They determine the likelihood of experiencing CS. For example, hand-based motion control leads to significantly less CS compared to whole-body motion control. These are some of the findings in research studies. Another aspect of locomotion that affects CS is the type of transition. So, locomotion means moving from one point to another.

One is how I control the movement; the other is how the scenes change, how the transition takes place from one field of view to another. So, when we are performing teleportation, that is an instantaneous transition, that type of transition seems to be less cyber-sickness-inducing compared to other means of locomotion. So teleportation is found to reduce cyber sickness compared to naturalistic free movement like walking, running, jumping, and so on. Field of view is another VR content-related factor that affects our cyber sickness susceptibility. What is field of view?

That is the extent to which the user's field of vision is captured by the display in a VR headset. Larger FOV means a larger field of vision is captured, and larger FOV generally results in greater immersion. Research has found that reducing the field of view can be an effective way to alleviate symptoms of cyber sickness. However, an excessively narrow field of view can lead to decreased immersion, which in turn leads to an increase in head movement to basically look for things in the environment, and that increase in head movement may be likely to exacerbate or increase CS.

Of course, it is a challenge to find the optimal FOV, but that is very crucial for creating a comfortable VR experience. Type of content is another important factor. Action content, fast movements, and abrupt alterations in the VR environment are found to induce more cyber sickness as compared to neutral and static content. This somewhat is quite obvious. However, users with high preference for adrenaline sports, those who like to take part in adventurous things and high adrenaline sports,

for them exposure to such content may not lead to cyber sickness. They are already accustomed to such behavior and such environment. So, for them, exposure to action content, fast movements, or abrupt alterations in the virtual environment may not lead to severe CS as compared to those who are not exposed to such types of sports or adventures. Genre of virtual environments also plays a significant role.

Content horror- content based on horror scenes is likely to induce more sickness as compared to pleasant general. Geometry and terrain of VR also affect CS. So, when in a virtual world a user is traveling in the presence of regular bumps and irregular terrain, then in such scenario CS is likely to happen as compared to when the same travel is taking place

with a flat terrain, on a flat terrain without bumpy rides. Weather conditions in the virtual environment: the kind of weather condition the user is exposed to can again affect the induction of CS.

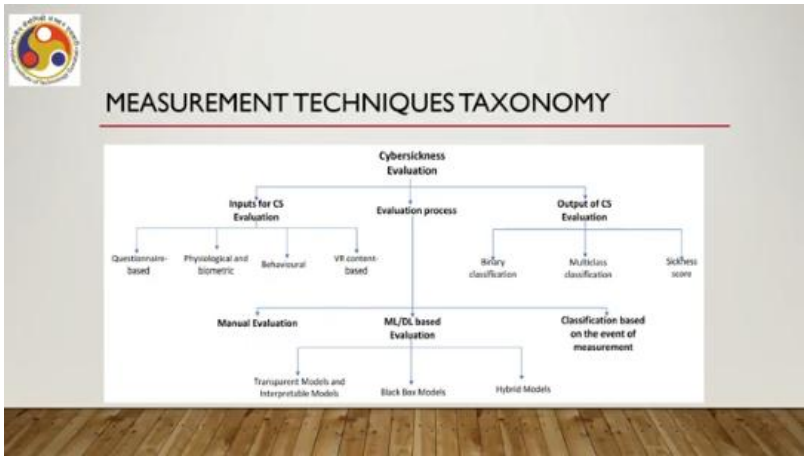
Higher CS is likely to happen during stormy conditions in the virtual environment as compared to clear weather conditions. So, these are obvious things, quite intuitive, and it has been found to take place in virtual environments. Duration of exposure related to VR content is another important factor. Several studies highlighted the importance of VR exposure time or duration on cyber sickness. Several studies were conducted.

Prolonged exposure was found to lead to greater CS when the exposure is through HMDs. That is, if users are wearing head mounted displays for a long duration, then cyber sickness is likely to be more. CS also increases with VR exposure time when users use different locomotion controllers during navigation tasks. Depending on the locomotion technique adapted to move from one point to another in the virtual environment, we have already seen that affects CS and that gets further complicated with longer duration.

Continuous exposure to live action 180 degree videos results in higher CS. This again increases with exposure time. Visual complexity: another external factor. Studies have shown that visual complexity is positively correlated to CS symptoms. So, we can positively correlate CS symptoms with visual complexity.

Increase in complexity leads to increase in CS. So, the more complex the scene is, comprising more objects of different shapes, sizes, and different movements, then the CS induction is likely to be more. So, that is about causes. So, earlier we covered internal factors and partly the external factors, namely related to hardware. In this part of the lecture, we have covered factors related to VR content which contribute to the onset of CS or severity of CS.

Now, in our next part of the discussion, we will cover: how do I know or how do I measure the extent of cyber sickness? How do I evaluate cyber sickness? Now, measurement techniques can be many, and the techniques rely on three things. One is the input: what input we are providing for measurement, then the measurement technique itself, and what kind of output we are generating.



There can be many forms of input: questionnaire-based, physiological and biometric information, behavioral information, content-based information. The evaluation process again, or the measurement techniques, can be different. We can have manual evaluation. Automated evaluation, machine learning-based, and real-time evaluation while the experience is taking place. There can be further subcategories also, as shown here in this summary.

The output also can be of different natures: it can be binary classes, multiple classes, or sickness score - a number. Let us go through each of these categories and subcategories in some detail to get a better understanding of the evaluation process. Let us start with the first thing, that is input. What input do we assume for measuring cybersickness?

So, there can be different forms of input. One is questionnaire-based. This is one of the common ways of taking input to measure cybersickness. So, these are self-report questionnaires or scales to rate users' discomfort or disorientation on a numerical scale. So, these are generally in the form of ratings.

Now, these types of rating scales for measuring CS are quite common. They are easy to administer and help the researchers or practitioners to better understand the causes and identify potential interventions. Typically, these scales are administered before or after the experience. Sometimes they can be administered even during the experience as well.

So, before, during, and after all three are possible, but mostly before or after and occasionally during the experience. So, one common scale of measurement used for cyber sickness is SSQ which stands for Simulator Sickness Questionnaire. It is in fact one of the most extensively used questionnaires for the purpose. It contains 16 items or

questions divided into three subscales: oculomotor subscale, disorientation subscale, and nausea subscale. Participants are asked to rate the intensity of individual symptoms. belonging to these three subscales, oculomotor, disorientation, and nausea, on a scale of 0 to 3, where a higher score denotes more severe symptoms. Now, SSQ was not originally designed for VR; it was designed for some other purpose, but adopted or rather adapted to use in the VR context, but there are other questionnaires that are specifically designed for VR, say, the Virtual Reality Sickness Questionnaire or VRSQ, then

we have the Cyber Sickness Questionnaire CSQ; this is good for evaluating VR experiences with HMDs. The second one, Then we have shorter and more concise questionnaires to assess specific conditions, such as Misery Scale or MIS, Fast Motion Sickness Scale or FMS. There are questionnaires for measuring susceptibility to CS, like MHQ or Motion History Questionnaire. MS-SQ or Motion Sickness Susceptibility Questionnaire, VIMS or Visually Induced Motion Sickness Susceptibility Questionnaire, VIMS-SQ. Also, there are some commonly used questionnaires to assess overall discomfort, namely Nausea Scale,

Motion Sickness Assessment Questionnaire or MSAQ. So, there are many such questionnaires that are available which are used to either detect or measure the susceptibility of cyber sickness. Another kind of input that we can consider for measurement apart from these questionnaires is the physiological or biometric input. Now, subjective questionnaires or these questionnaire-based inputs can be subjective and which are prone to inaccuracies. It may not always be reliable.

So, to overcome that we can use physiological inputs or biometric inputs which can be used to detect CS. Such inputs can be heart rate, galvanic skin response, EEG, and so on. There are in fact a large number of such inputs which we can use. Let us see what kind of inputs we are talking about here. EEG or electroencephalogram, it is one of the widely used inputs to detect CS.

Then we have electrocardiogram ECG, then electrogastrogram EGG, electrooculography EOG, respiration signals, photoplethysmography, PPG and electromyogram or EMG. Also, we can make use of galvanic skin response or GSR. So, that is about different sorts of physiological or biometric inputs. Another kind of input can be behavioral input.

It is possible to detect CS using behavioral cues as well, such as blink rate, head and body movement, how we move them, and postural sway. These can have useful features for measuring CS. Axial movements or changes in the user's center of pressure during VR

exposure can be used to monitor postural sway, which is one of the ways to measure CS. Studies have shown a positive correlation between postural instability and the severity of CS. So, postural sway can be measured using axial movements, and it has been found that postural sway

can be an indicator of the severity of CS. Content-based input can also be taken to measure CS, to evaluate CS. The complexity of the visual environment in VR can contribute to CS, as we have already seen earlier as part of the cause. Now, researchers have studied different features related to VR content to evaluate CS. What are those features that stand out for CS measurement?

Optical flow, which is the apparent motion perceived by VR users due to relative motion between the user and the environment. Depth cues, exceptional motion velocity, which can be another key cause of CS. So, these are some of the content-based inputs that can be utilized to measure CS. So, we have covered the inputs part, inputs for CS measurement. There are four such categories of inputs: questionnaire-based, which is subjective; physiological and biometric, objective; then behavioral and content-based.

Now let us have a quick look at the evaluation process. What kind of techniques can we use to measure CS based on the input? Now these measurement techniques can broadly be of two types. One is manual evaluation that is based on human intervention. The other one is machine learning-based evaluation or automated evaluation, where automated algorithms are used,

on the input to measure CS. Typically, assessment conducted by means of questionnaire such as SSQ or VRSQ comes under manual evaluation. It is quite obvious. So, there somebody has to distribute the questionnaire, collect the feedback and go through the responses to come up with a decision on the severity or onset or possibility of cyber sickness.

So, that is entirely human intervention to measure and that comes under manual evaluation. As opposed to that, we can make use of other inputs, say for example, the psychological, physiological or biometric input to train models to automatically determine cyber sickness. So, that type of automated approach comes under ML based evaluation. Now, there can be different ways to perform ML based evaluation.

One way is through the use of transparent models. So, these types of models provide clear insight into the decision-making process and the model's inner workings can easily be

understood. So, when we are able to do that while performing an automated or machine learning-based evaluation process, then we say that we are making use of transparent models. Many studies have reported transparent model-based approaches to major sickness from objective or physiological or biometric-based inputs.

For example, stepwise regression analysis: stepwise regression analysis is utilized to measure CS using a variety of physiological signals, physiological inputs. Now, stepwise regression can be considered to be a transparent model. So, we have covered a part of evaluation strategies. In this part of the lecture and the previous part, we have covered the causes, what are the things that lead to cyber-sickness, that lead to cyber-sickness.

Also, in this lecture, partly we have covered evaluation. We have seen that there are three aspects of evaluation or measurement. input: what kind of input and how we are getting the input; then the process: how we are processing the input; and the third one is output: what we are getting as output On the input front, we have seen that we can go for subjective questionnaire-based rating scales, rating scores as input, or we can go for objective inputs such as physiological signals or biometric signals.

or we can take behavioral input or content-based input. In the evaluation part, we have seen that we can go for manual evaluation, typically based on questionnaires or rating scores, or we can go for automated evaluation, typically using machine learning models. Now, we started our discussion with transparent models. These are models that are easy to understand, and we know how they work.

The remaining part of the evaluation, as well as the mitigation strategy, we will cover in the later part of this lecture. So, I hope you have understood the concepts and look forward to meeting you all in the next part of the lecture. Until then, thank you and goodbye. Happy learning.